Lab 11

Poly Phase Systems

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11.1 Objective

To analyze poly phase systems with balanced and unbalanced loads connected in wye or star configurations using MATLAB Simulink.

Note: Attach snippet of all the Simulink models and graphs, all the graphs and circuits must be labelled properly along with the task and headers.

11.2 Introduction

Polyphase systems are electrical power distribution systems that use multiple alternating current (AC) voltages, typically with different phases, to transmit and distribute electrical energy. These systems are commonly used for power generation, transmission, and distribution in large-scale electrical networks due to their efficiency and ability to handle higher power loads.

In a polyphase system, the AC voltages are generated with a specific phase difference between them. The most common types of polyphase systems are three-phase and two-phase systems, we will be working on three phase systems in this Lab.

11.3 Three Phase System

In a three-phase system, three AC voltages are generated, each with a 120-degree phase difference from the others. These three phases are typically denoted as phase A, phase B, and phase C. The combination of these three phases creates a balanced three-phase system, where the load is evenly distributed among the phases, resulting in smoother power delivery and reduced voltage fluctuations.

Let's understand it through a practical example, single phase AC generator produces a single sinusoidal voltage for each rotation of rotor. If number of coils on the rotor is increased in a specified manner, the generator will develop more than one AC phase voltage per rotation of rotor and is called poly-phase generator. A three-phase generator generates three sinusoidal voltages each having a phase displacement of 120^o. This is achieved by placing induction coils 120^o apart on stator. Each phase has equal frequency. It is equivalent to connecting three single phase sources together.

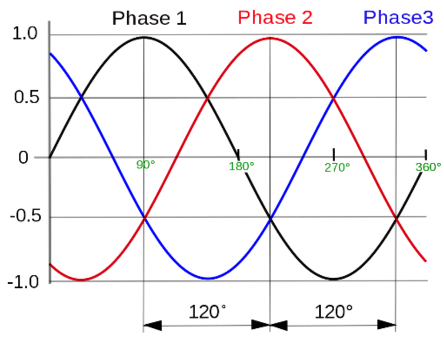


Figure 1: Waveform of Three-Phase Voltages

11.3.1. Wye and Delta connection

In three phase systems, there are two connections: Wye or Star connection and Delta connection. The common point in Y connection is called neutral or star point. Based on the connection of source and load in different configurations, there are following 4 possible combinations:

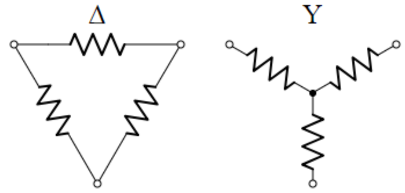


Figure 2: Wye and Delta Connections

a) Y-Y connection
b) Y-D connection
c) D-Y connection
d) D-D connection

11.3.2 Balanced and Unbalanced Source and Load

A three-phase source is said to be balanced if the voltage per phase is equal and the three phases have phase difference of 120^o. A load is said to be balanced if impedance per phase is equal.

11.3.3 Line and Phase Quantities

In a three-phase system, line voltage refers to the voltage measured between any two-line conductors while phase voltage refers to the voltage measured across the two ends of a component.

In Y-connection, phase voltage will be the voltage of any phase conductor with respect to neutral (Van, Vbn, Vcn) and line voltage (also called line to line voltage) is the voltage between two phases (Vab, Vbc, Vca). In delta connection, line and phase voltages are equal and for Y-connection, the relation between line and phase voltages is given below.

$$V\ line = \sqrt{3}\, Vphase$$

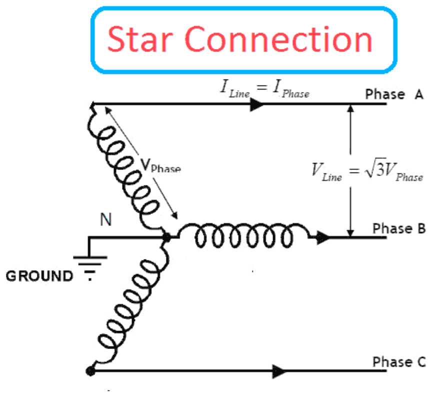


Figure 3: Delta and Wye Systems Schematic

Line current is the current flowing through the line conductors (Ia, Ib, Ic) and phase current is the current through any component. In Y-connection, line and phase currents are equal. For delta connection, the relation between line and phase currents is given by: $I\ line = \sqrt{3}\ Iphase$

Neutral current

The current flowing through neutral is the phasor sum of all phase currents. For a balanced network, this will be equal to zero.

$$In = Ia + Ib + Ic$$

Per phase analysis and Wye-delta conversion

A three-phase system can be easily analyzed using per phase analysis in which each phase can be seen as a single-phase system when its phase quantities are known i.e. phase voltage and per phase impedance. For Y-Delta source and load configuration, the load connected in delta can be converted into its wye equivalent using equation given below:

$$Zd = 3\,Zy$$

11.4 Introduction to Simulink

Simulink is a visual programming interface designed to make modelling systems intuitive. Simulink is an extension to MATLAB that allows engineers to rapidly and accurately build computer models of dynamical systems using block diagram notation. It offers a way to solve equations numerically using a graphical user interface, rather than requiring code. Models contain mainly blocks and signals

- Blocks are mathematical functions; they can have varying numbers of inputs and outputs.
- Signals are lines connecting blocks, transferring values between them. Signals are different data types, for example numbers, vectors or matrices. Signals can be labelled.

11.4.1 Exploring Simulink

1) To start the Simulink software, first start the MATLAB® technical computing environment.
2) To start Simulink Library Browser from MATLAB, use one of these approaches:
3) On the MATLAB toolbar, click the Simulink button

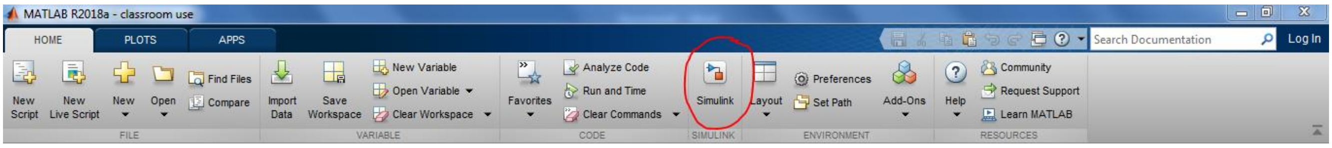


Figure 4

4) At the MATLAB prompt, enter the Simulink command.
5) Select "*Blank Model*" and Simulink Editor window will be opened.
6) Save this model. Go to *File> Save.*
7) There are different options in menu bar and tool bar at the top and in palette at the right side.

8) Open Library Browser by clicking on the icon in menu bar or select *to View > Library Browser*. There are different libraries in Simulink. The library needed to simulate electrical power systems is *"Specialized Power Systems"* which can be found under "*Simscape/Electrical*" library.

There are different categories of blocks including *Electrical Sources*, *Elements*, *Measurements*. Open these to explore the blocks available in these. There is a **powergui** block present in *Specialized Power Systems*. This is used to set simulation type and parameters and is required to run simulation of other fundamental blocks. Refer figure below:

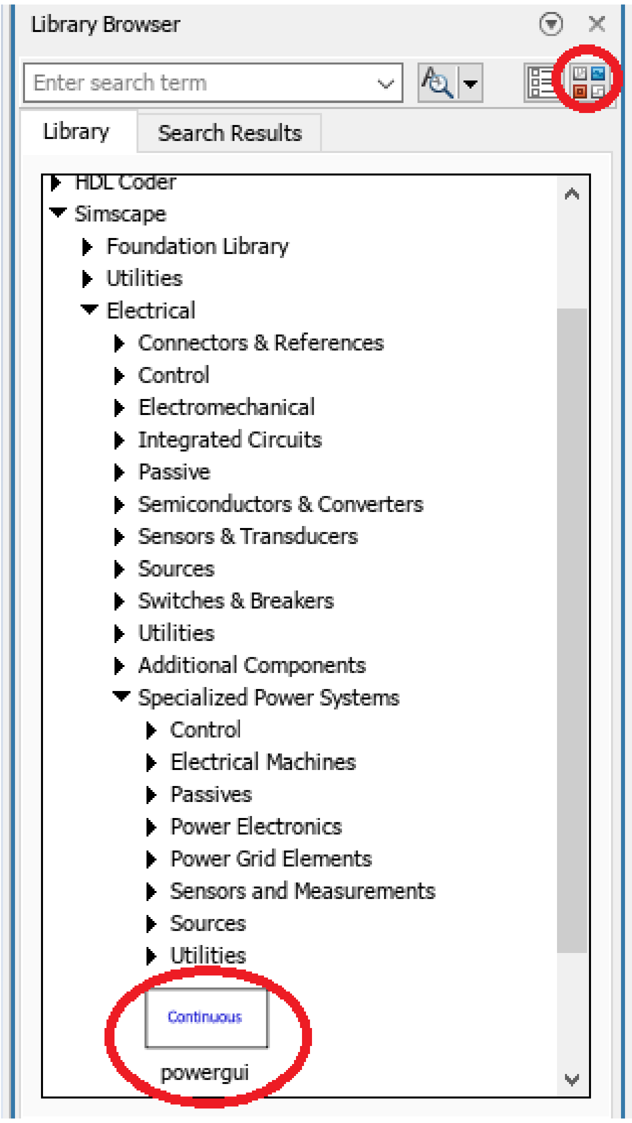


Figure 5: Power gui block and click launch standalone Library icon

9) To add any model, you can just select it and drag it into the editor or *Right Click> Add block to model*
10) Each block has its own settings in Block Parameters and Block Properties.

Task 1: To simulate three-phase Y-connected balanced source

Attach all tasks Simulink model and graphs with manual

1) Add three single phase AC voltage sources from **Simscape** → **Electrical** → **Specialized Power Systems** → **Sources** → **AC Voltage Source** in your model and connect them in Y configuration.
2) Open block parameters by double clicking the icon. Add peak amplitude of 100 V, frequency 50 Hz, Sample time: 0.0001. Add phase in degrees for each source 0, -120 and 120 and select measurement "Voltage" from drop down menu.
3) Connect the star point (common point in Y connection) with neutral. Neutral block can be found in **Simscape→Electrical→Specialized Power System→Power Grid Elements→Utilities.**

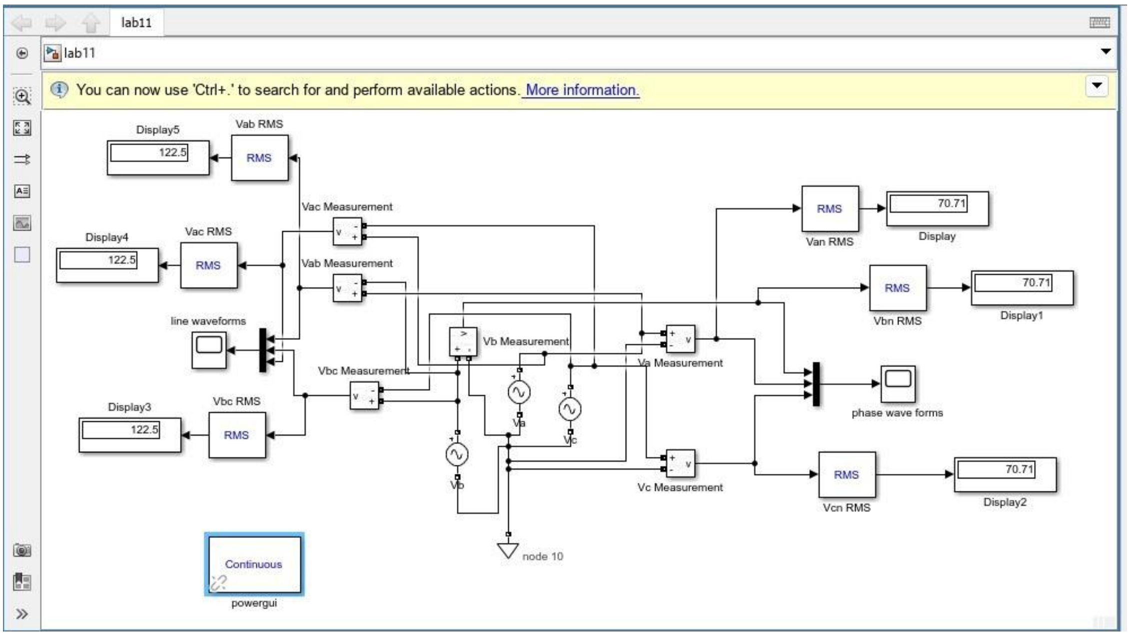
4) a) To measure the voltage, add *Voltage Measurement* block from **Specialized Power System → Sensor & Measurements → Voltage Measurement**. To measure instantaneous phase voltage, connect *Voltage Measurement* block across one phase i.e., source and neutral point.

   To display multiple waveforms in one plot, use ***mux*** with scope. Search for *Mux* or find it in *Simulink > Commonly Used Blocks*. Connect the output of all voltage measurement blocks to inputs of mux and then connect mux with Scope. You can change number of inputs to mux. *Scope* can also be found in the same *library*.

5) Run the simulation after adding ***powergui*** block to your model. Set simulation stop time equals to 0.1s in the tool bar. Double click on scope to see the waveforms.
6) Measure line voltages by connecting a *Voltage Measurement* block across two phases. Display all the line voltages in one scope.

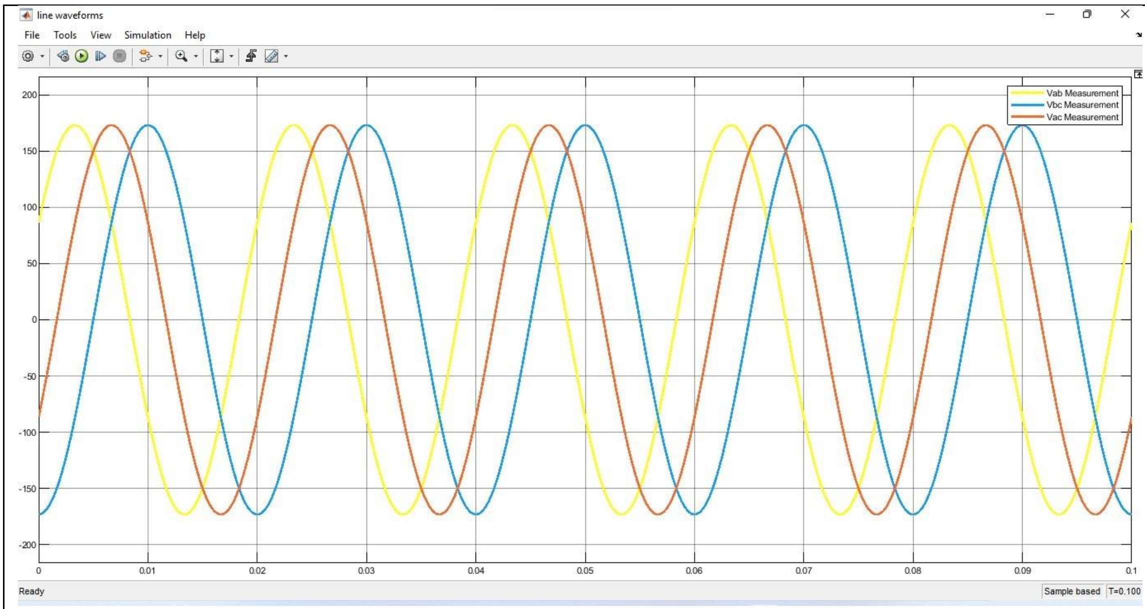
7) b) To measure RMS value of phase and line voltages, connect ***RMS*** block (written RMS in **blue** on block) from **Specialized Power System → Sensor & Measurements** with voltage measurement and connect ***Display*** block at its output. RMS block can be found in power system-specialized technology-control & measurements-measurement-RMS (set fundamental frequency to 50 Hz and Initial RMS value to 0) and *Display* in *Sinks*.
8) Observe and analyze the waveforms and record values of line and phase voltages obtained from display block. Also, attach the screenshot of Simulink Model and Waveforms obtained.

Attach your Simulink Model with report



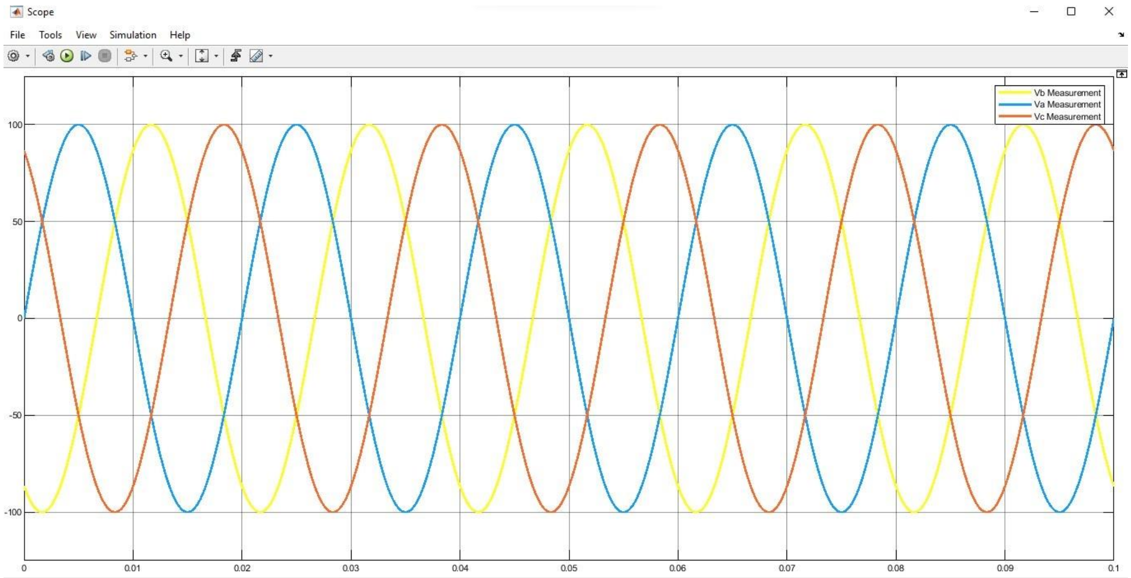
Attach plots of Line & Phase Voltages

Line:



Value= approx. 173V

Phase:



Value is approx 100V

Analysis:

In the waveforms we can see clearly that the phase and line voltages are 120 degrees apart from each other.

we can also see that vline is $\sqrt{3}$ vphase as vphase is 100 and v line is approx. 173. This tells us that the simulink model is correct as it satisfies this condition of Y connected sources.

Task 2: To measure voltage, current and power of three phase balanced Y-connected load.

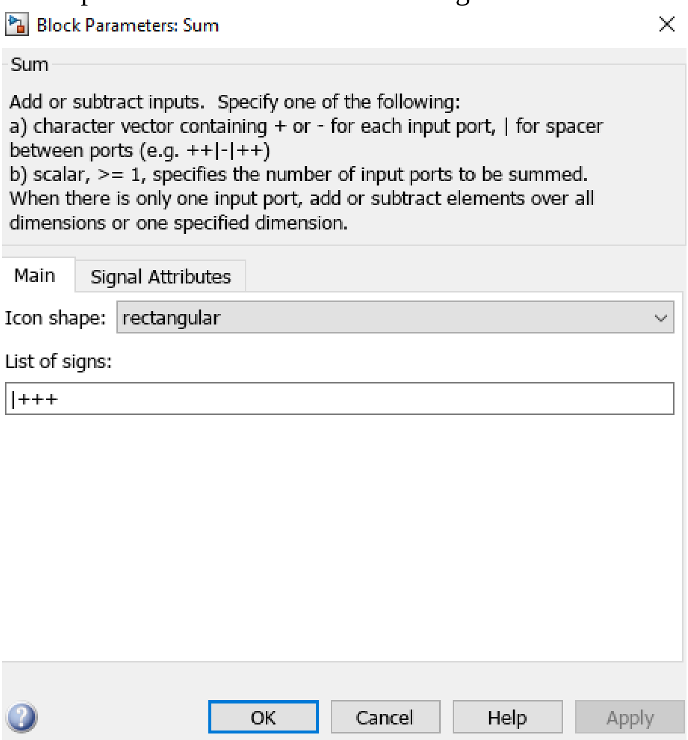
Attach all tasks Simulink model and graphs with manual

1) Add **series RLC branch** in your model from **Specialized Power System → Passive**. Connect such 3 branches in Y connection to simulate Y-connected load. set impedance per phase equal to 100 Ohms. (Select branch type to R only).

2) a) To measure instantaneous line currents, add *Current Measurement* block in series with AC voltage source and R load. Add scopes and display block with RMS to display the three-line currents. Attach model and plots.

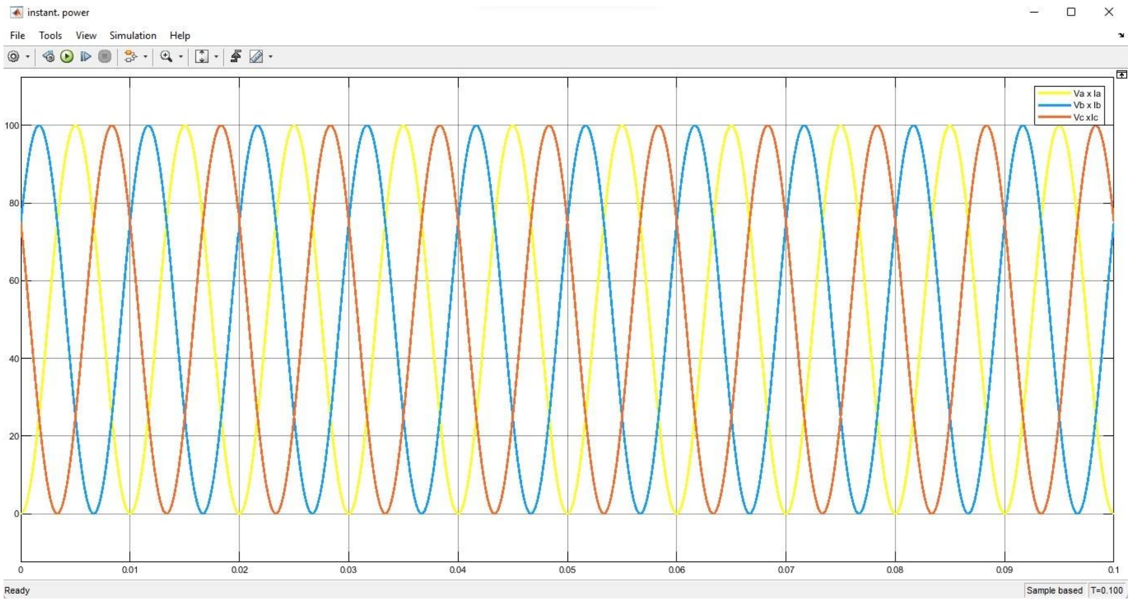
3) b) To plot instantaneous power per phase, add ***Product*** block from ***Simulink → Commonly Used Blocks***. Now using this find product of phase voltage waveform and phase currents waveform to get instantaneous power of *each* phase and see the waveforms using **scope**. In **scope** more channels can be added double click on scope go to file and set the Number of input port to 3. Attach model and plots.

4) c) To find total power, add power per phase using ***Sum*** block form Commonly Used Blocks and connect its output with Scope. To add 3 inputs double click on sum block and then you need to add one more "+" sign, the shape of the block can also be changed. Attach model and plots.



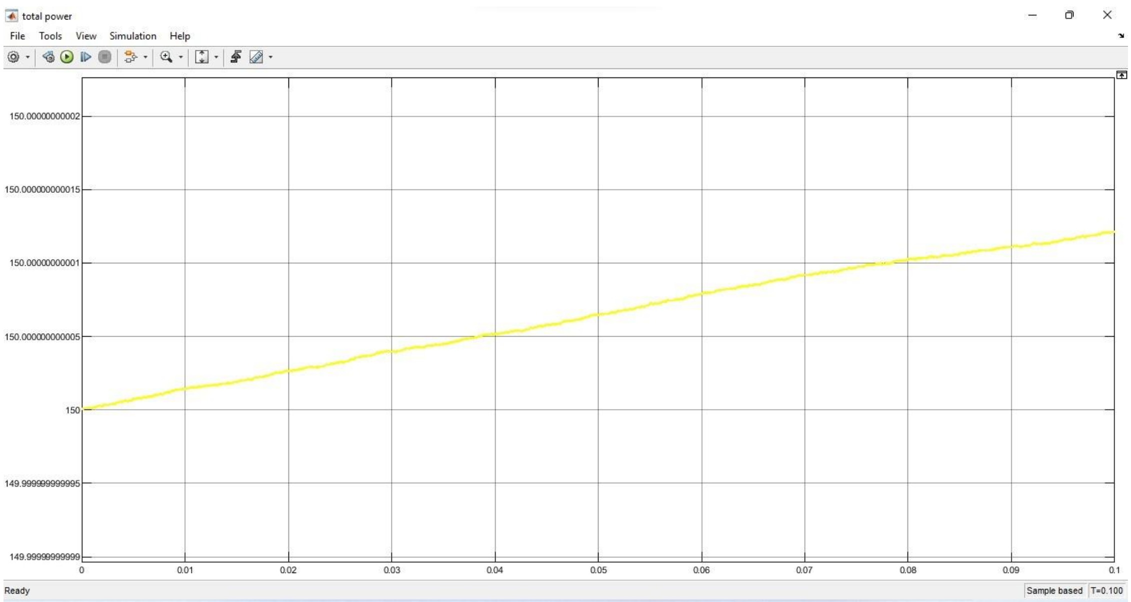
5) Analyze the waveforms of instantaneous power per phase and total power. Is total instantaneous power constant? Record the value and attach screen shot of waveform.

Instantaneous Power:



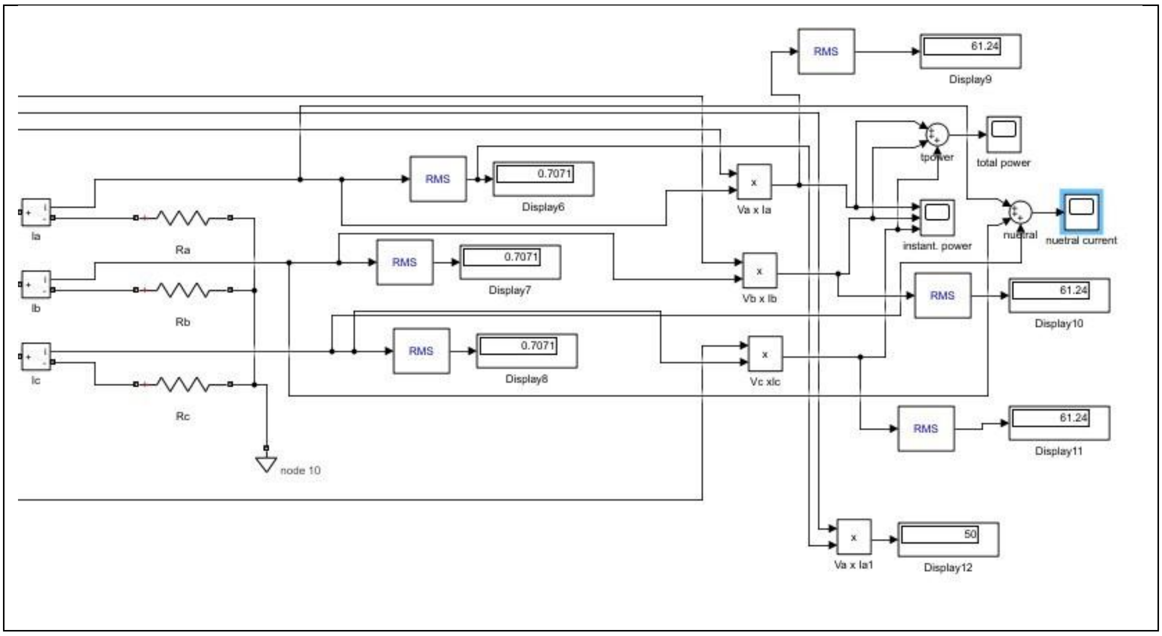
Value is 100W. The waveforms appear sinusoidal and balanced in terms of amplitude and frequency, with a consistent phase shift of 120 degrees between the phases. Showing a system with balanced load

Total power:



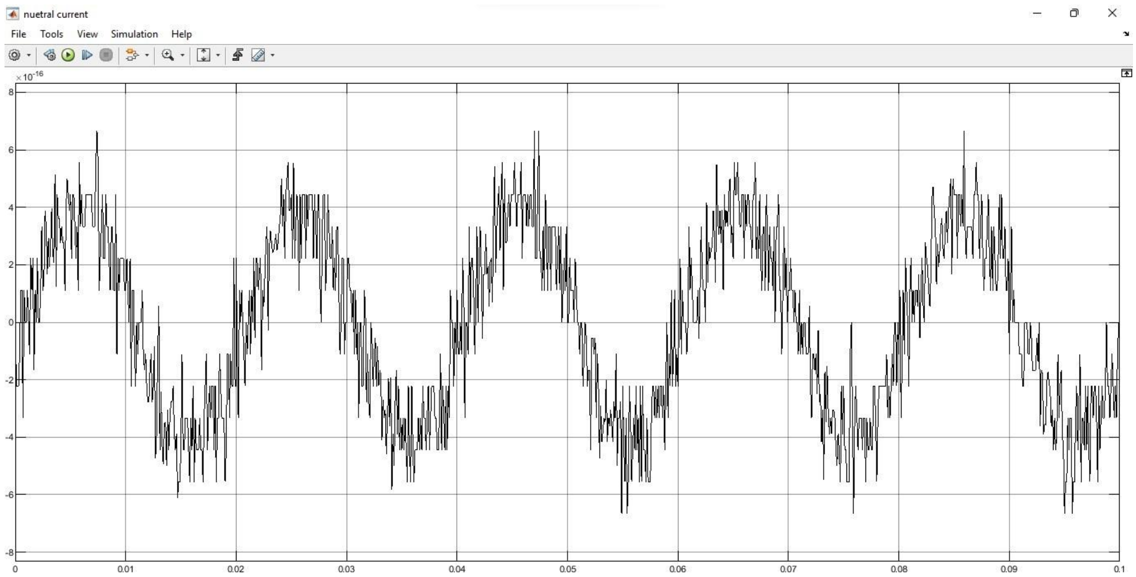
Value is 150W. Yes we can say that the total power is constant even though it seems as if it is increasing, it is increasing at such a small rate that it is negligible and can be said to be constant.

Simulink model:



6) d) Measure and record neutral current. Is neutral current equal to zero? Attach model and plots.

Neutral current:

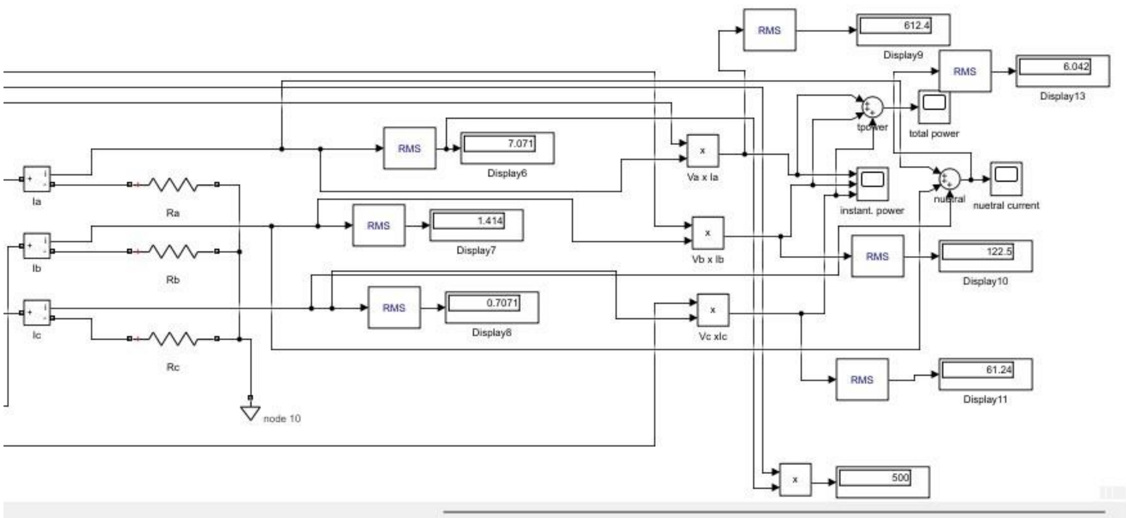


Based on the plot, we see that the neutral current is fluctuating around zero, but it is not exactly zero. The neutral current is almost negligible but not perfectly zero. There might be some minor imperfections causing minimal current flow through the neutral wire

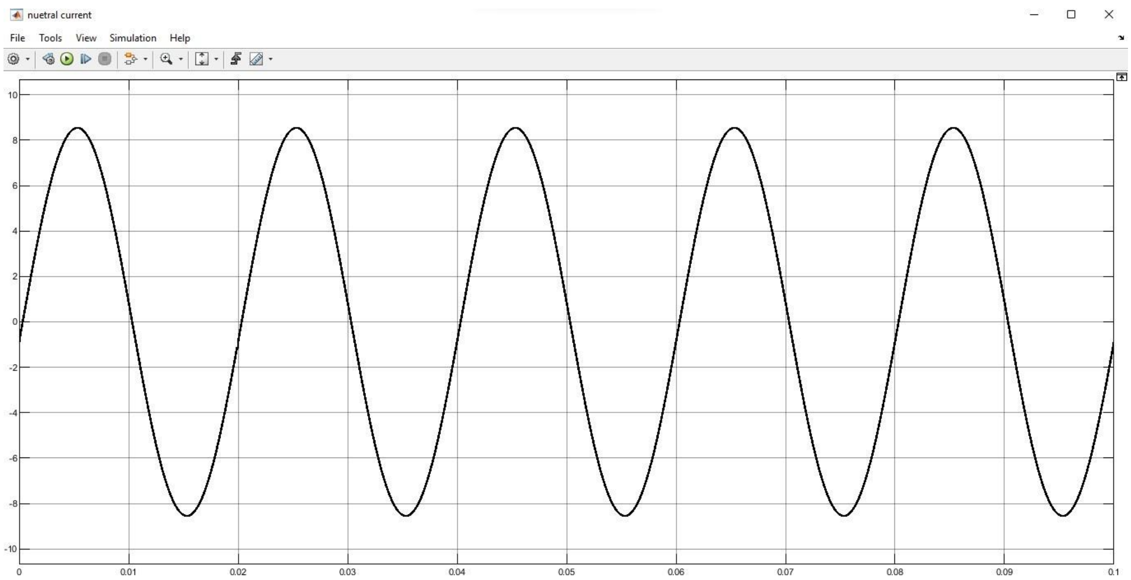
Task 3: To measure neutral current for unbalanced load

1) Change the load impedance per phase for a Y-connected load such that Ra =10 Ohms, Rb=50 Ohms and Rc=100 Ohms.
2) Measure and record neutral current. Note down your observations below:

Simulink model:



Neutral current:



The sinusoidal neutral current indicates an unbalanced load in the Y configuration. Due to unequal currents in each phase, the imbalance results in a residual current flowing through the neutral wire.

3) Attach all tasks Simulink model and graphs with manual.

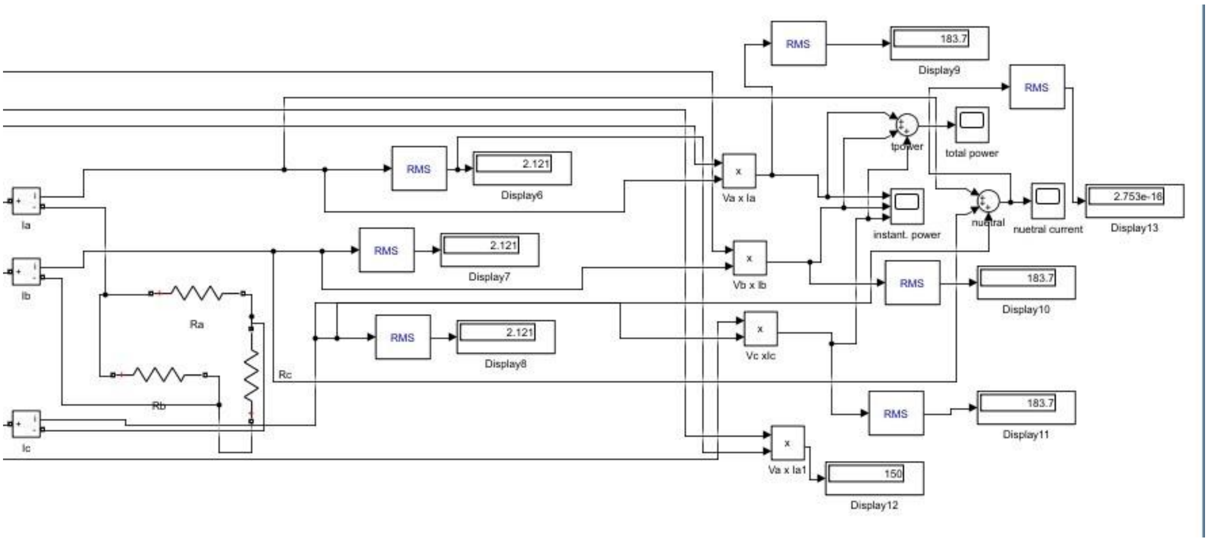
Post-Lab:

Task 4: To measure voltage, current and power of three phase delta-connected load

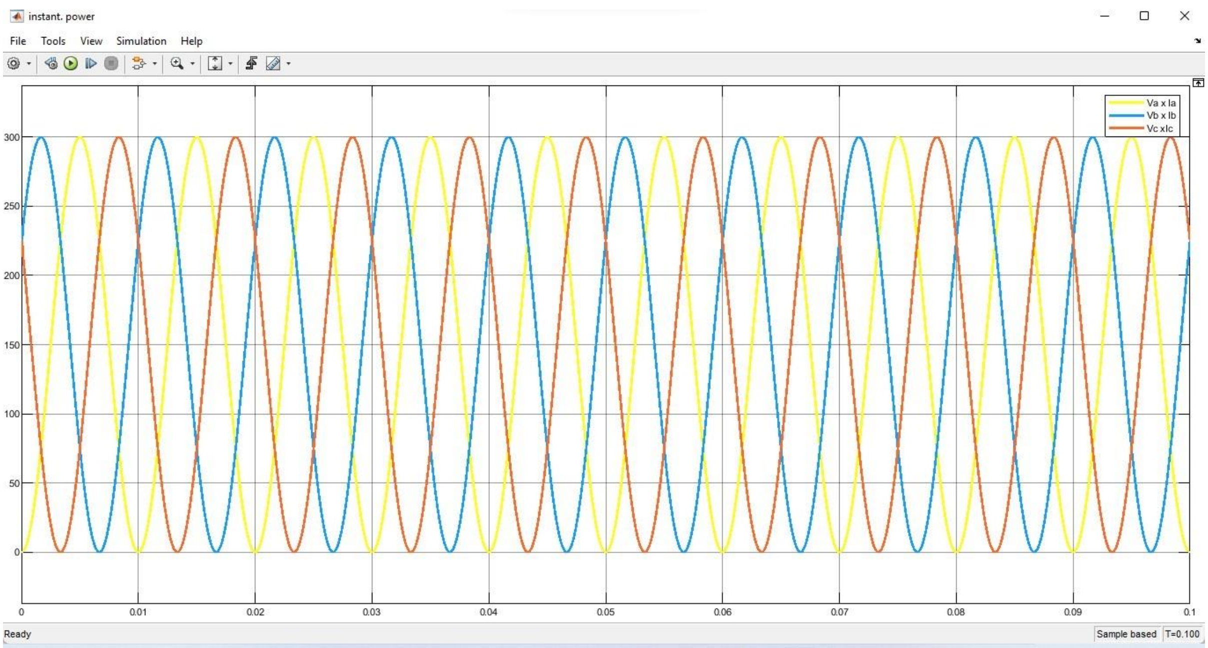
Connect a three-phase balanced load in delta connection with per phase load of R= 100 Ohms.

Measure instantaneous Line and phase Voltage, Current and Power per phase. Also, measure total 3 phase power & neutral current. Show all Peak & RMS values and attach screenshot of Simulink model & waveforms in report. Provide you analysis on attached graphs.

Simulink model:

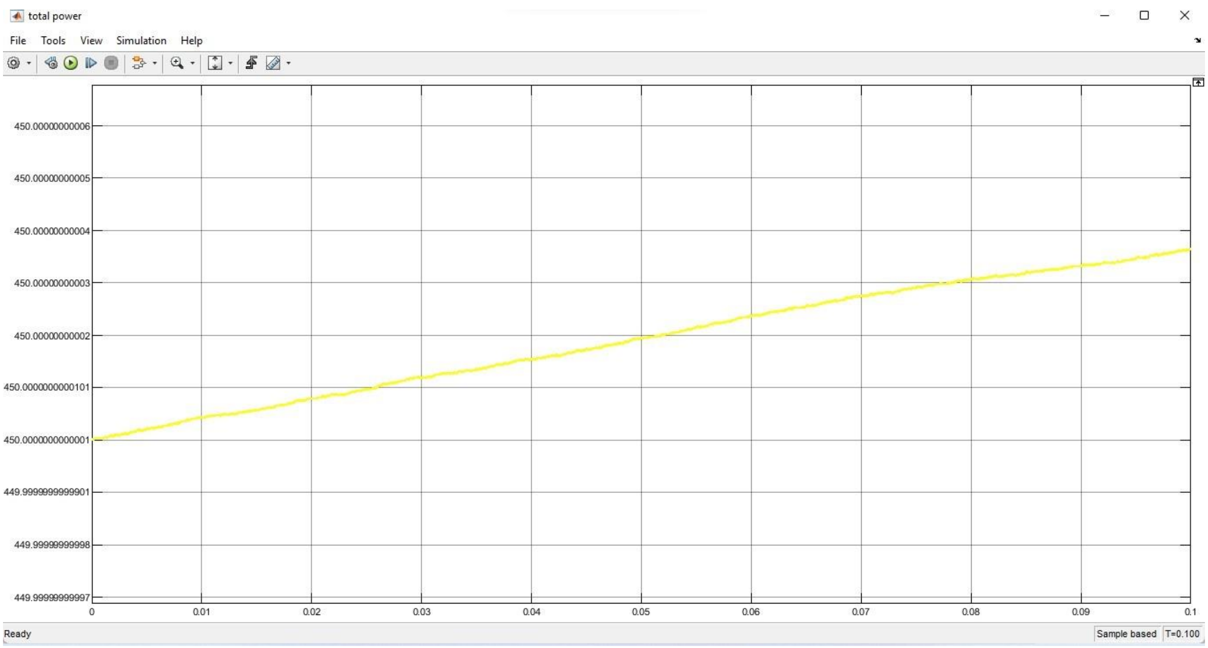


Instantaneous Power:



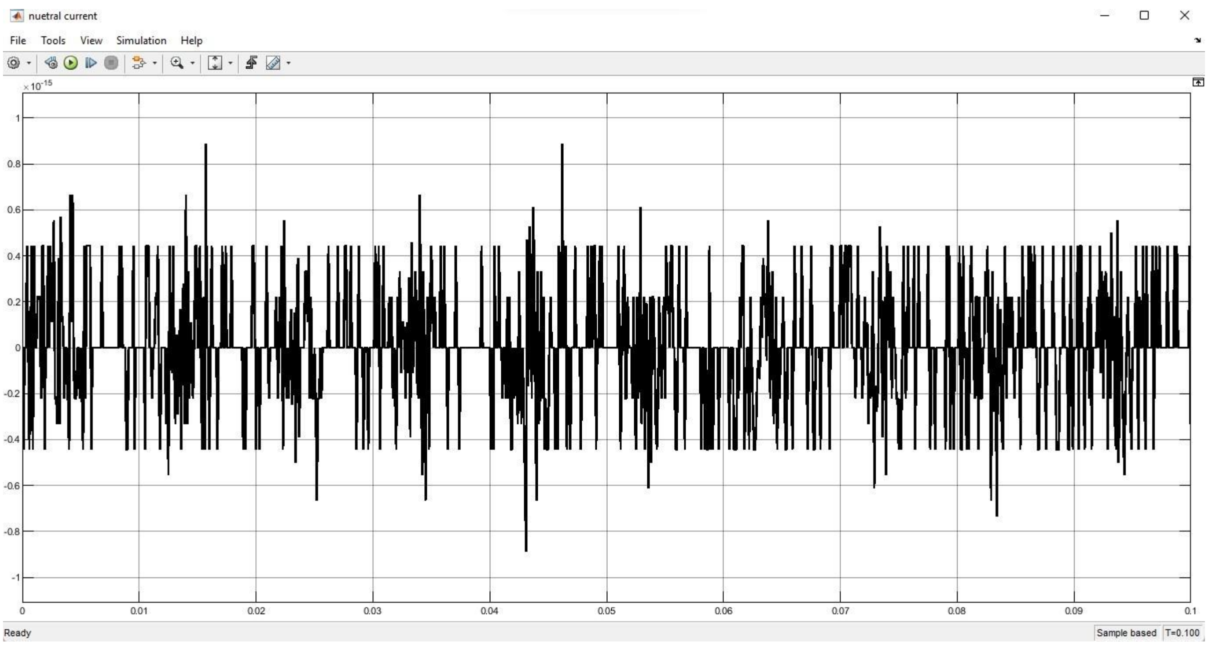
Value is 300W. The waveforms appear sinusoidal and balanced in terms of amplitude and frequency, with a consistent phase shift of 120 degrees between the phases. Showing a system with balanced load.

Total power:



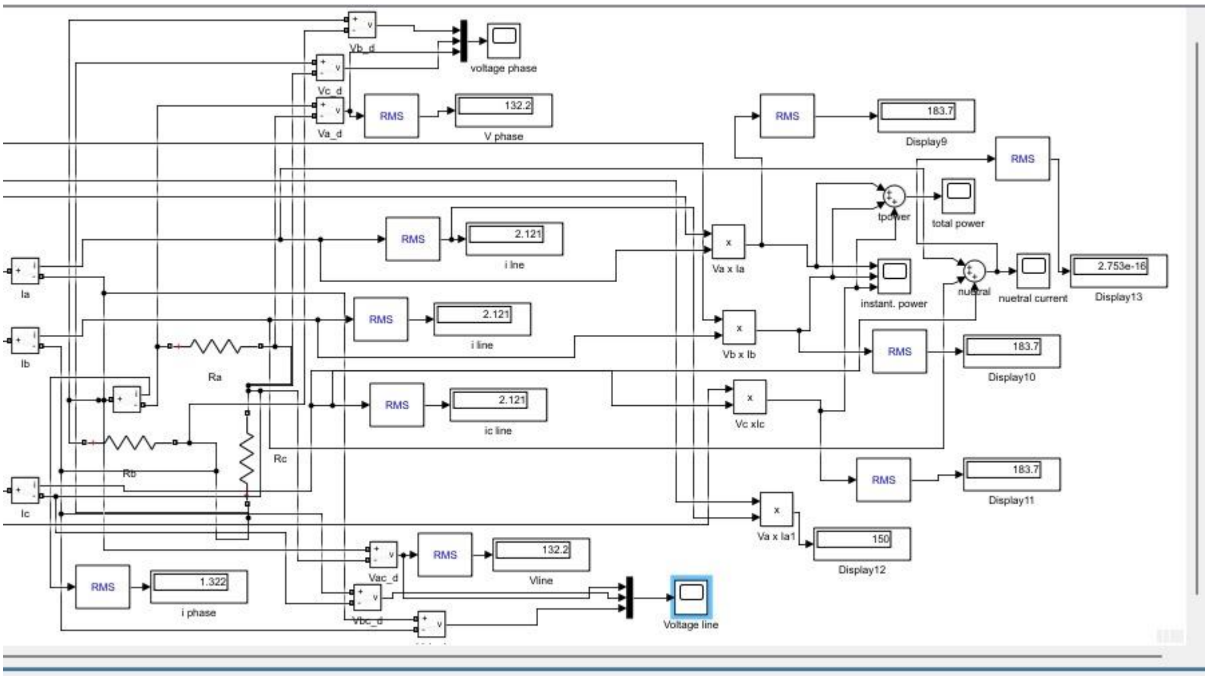
Value is 450W. Yes we can say that the total power is constant even though it seems as if it is increasing, it is increasing at such a small rate that it is negligible and can be said to be constant.

Neutral current:

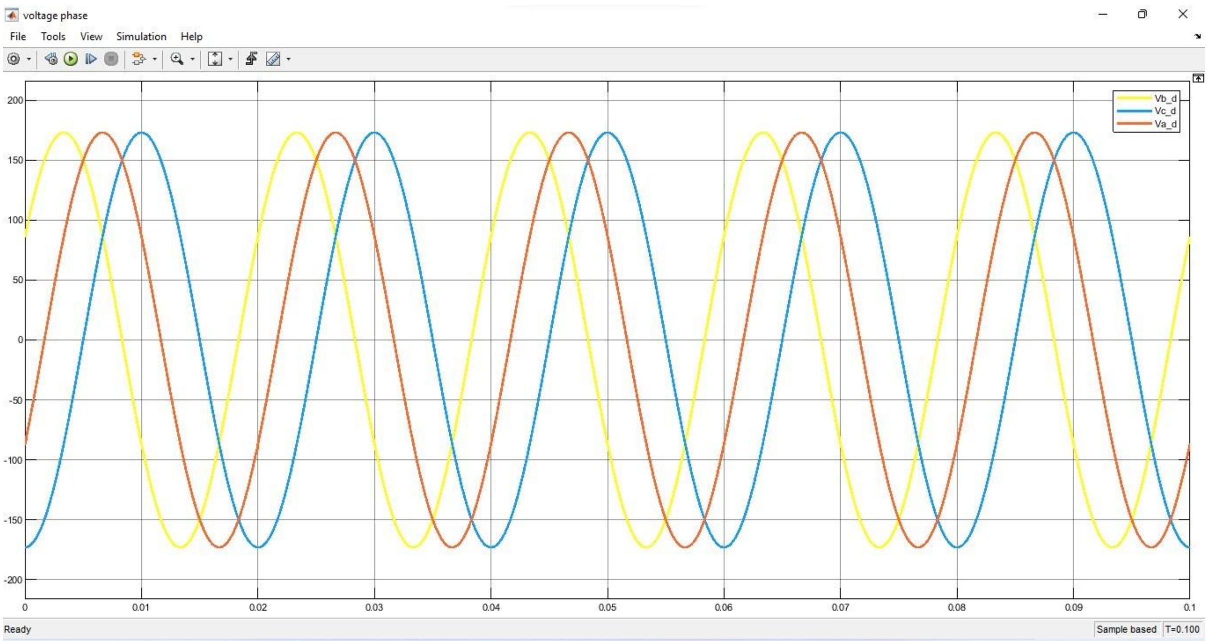


Based on the plot, we see that the neutral current is fluctuating around zero, but it is not exactly zero. The neutral current is almost negligible but not perfectly zero. There might be some minor imperfections causing minimal current flow through the neutral wire.

Simulink model to show voltage:

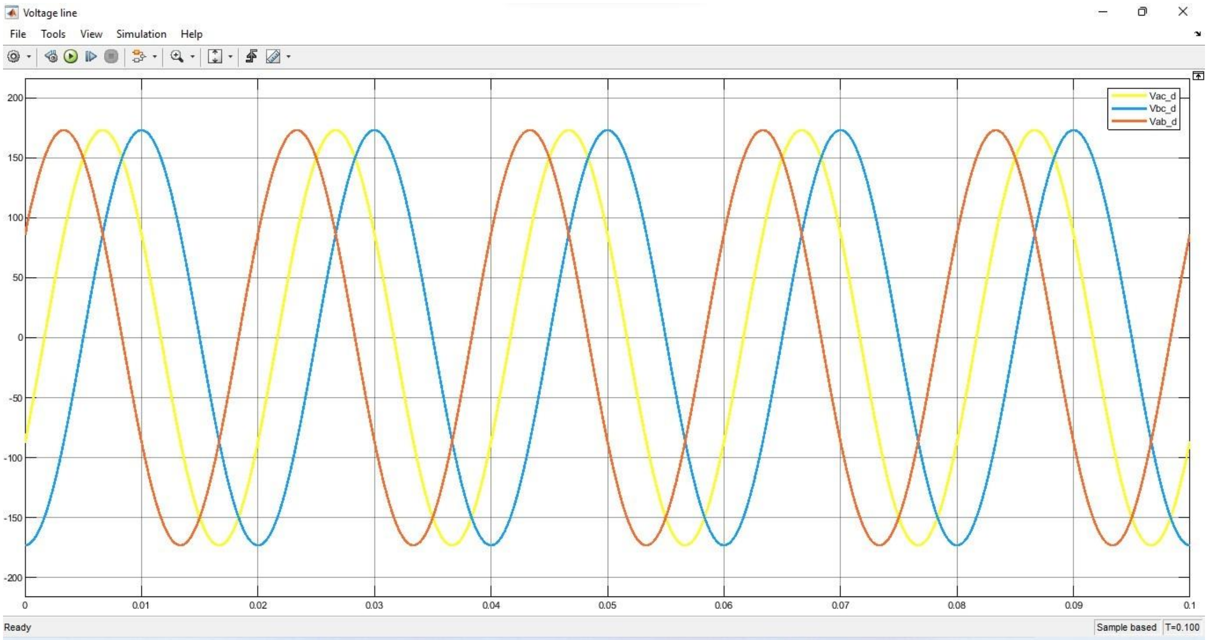


Vphase:



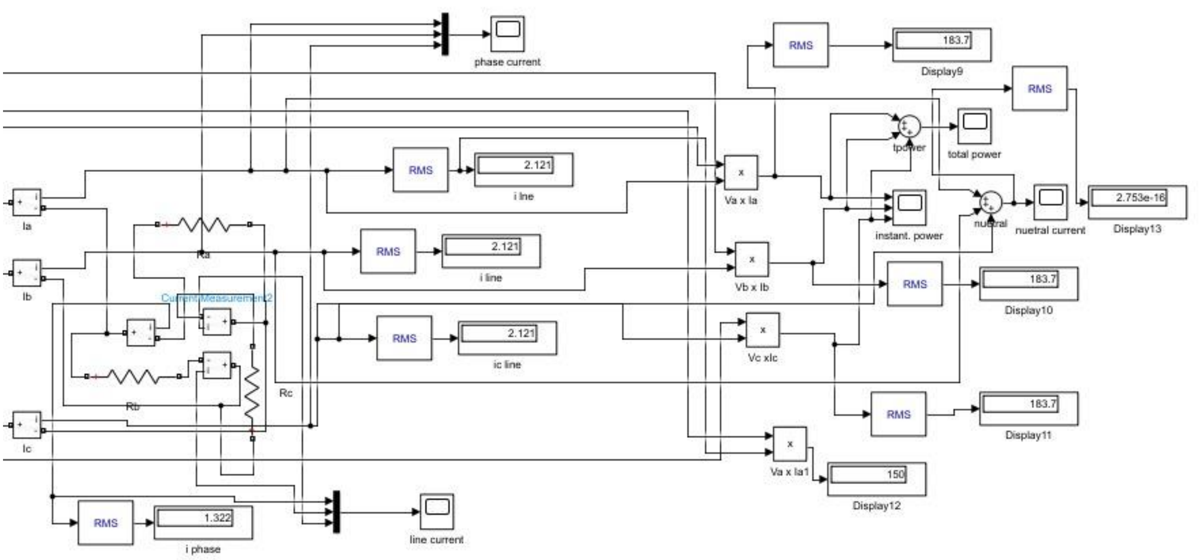
Value is 173V.

Vline:

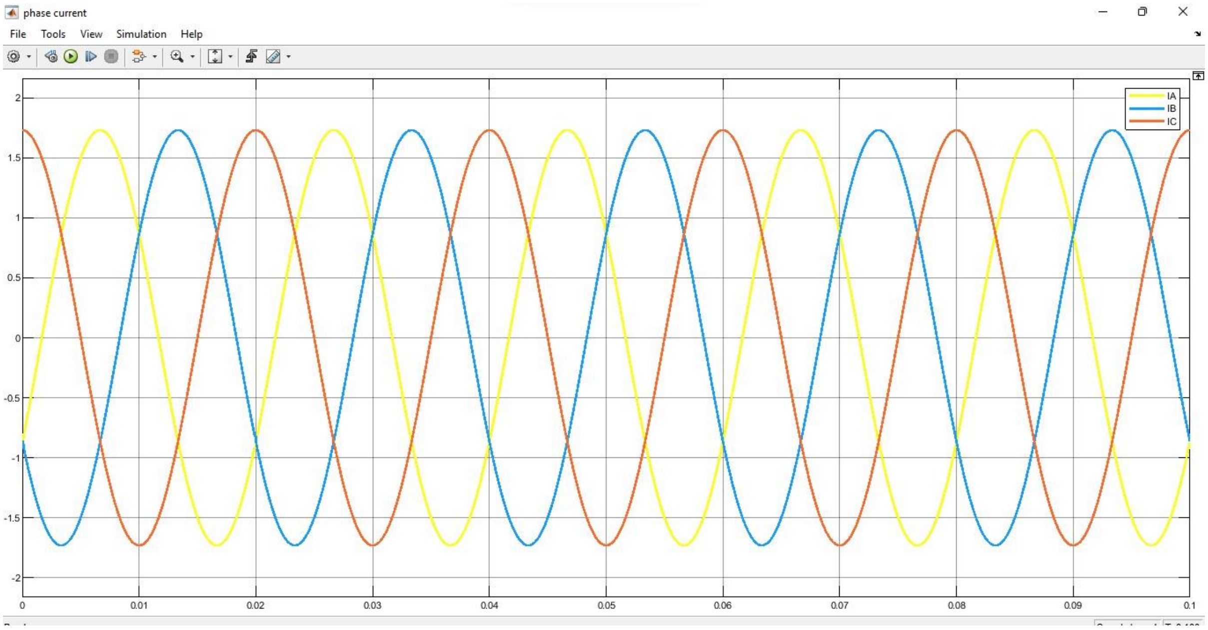


Value is 173V. As we can clearly see from the graphs above vline = vphase this satisfies the condition of load connected in delta configuration. We can also see that both graph's waveforms are at a phase shift of 120 degrees.

Simulink model to show current:

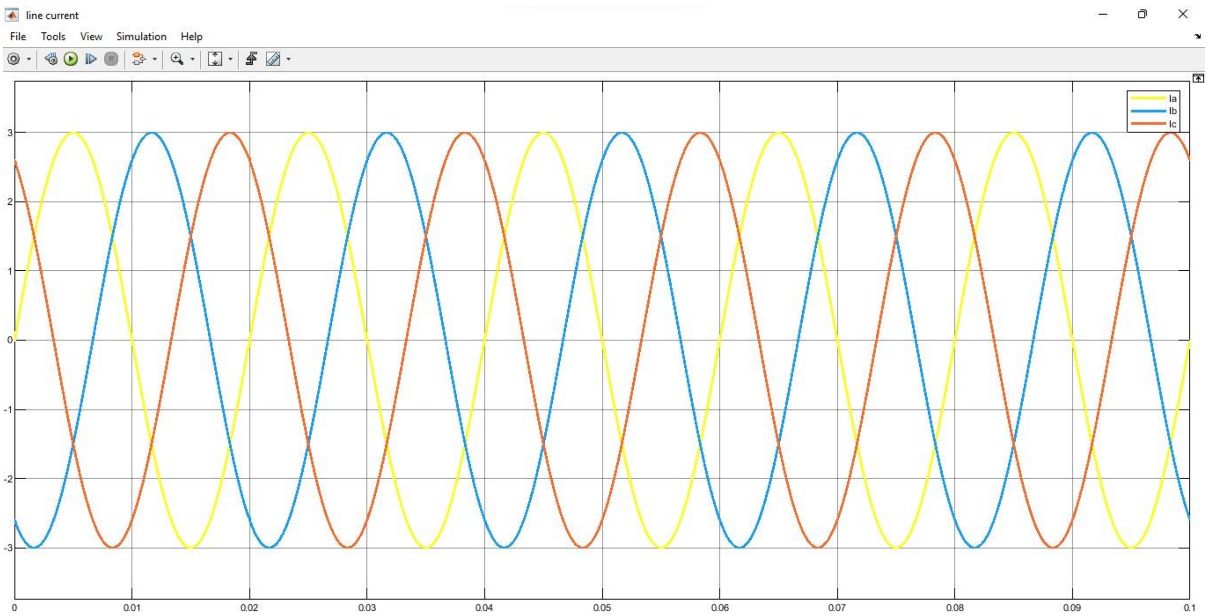


Iphase:



Value is 1.73A.

Iline:



Value is 3A. As we can clearly see from the graphs above iline = $\sqrt{3}$ iphase this satisfies the condition of load connected in delta configuration. We can also see that both graph's waveforms are at a phase shift of 120 degrees.

Frequently Asked Questions (FAQs)

Q1. Why three-phase is preferred over single-phase?

Answer: Three-phase power, three-phase power supplies are more efficient. A three-phase power supply can transmit three times as much power as a single-phase power supply, while only needing one additional wire (that is, three wires instead of two).

Assessment Rubric
Lab 11
Poly Phase System

Name:	Student ID:

Points Distribution

Task No.	LR 2 Simulation	LR 4 Data Collection	LR5 Results/Plots	LR 6 Calculation
Task 1	/5	/5	/10	/5
Task 2	/5	/5	/10	/5
Task 3	-	/5	/5	-
Task 4	/10	/5	/10	/5
Total	/20	/20	/35	/15
CLO Mapped	CLO 3	CLO 3	CLO 3	CLO 3

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission	/10	CLO 4

CLO	Total Points	Points Obtained
3	90	
4	10	
Total	**100**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR1	**Circuit Layout**	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR2	**Program/Code/ Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	**Data Collection**	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	**Calculations**	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
LR10	**Analysis**	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	**Design**	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.